Robust Test-Time Adaptation in Dynamic Scenarios

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Summary

Problem Practical Test-time Adaptation (PTTA).

- Why consider distribution change and correlative sampling simultaneously?
- How to address erroneous normalization, local overfitting, error accumulation?

Contributions.

- We propose a new setup namely practical test-time adaptation (PTTA), which considers both distribution change and correlative sampling.
- We propose a robust test-time adaptation method (RoTTA), which has a more comprehensive consideration of PTTA challenges.
- RoTTA obtains state-of-the-art results on common TTA benchmarks and a large-scale DA dataset, and outperforms the best baseline by a large margin.

GitHub: https://github.com/BIT-DA/RoTTA

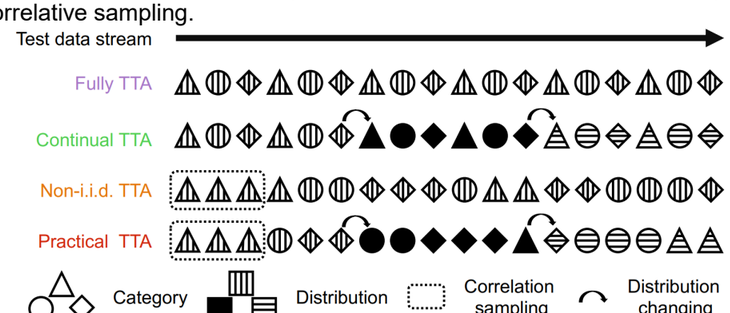
Motivation

Practical Test-time Adaptation (PTTA).

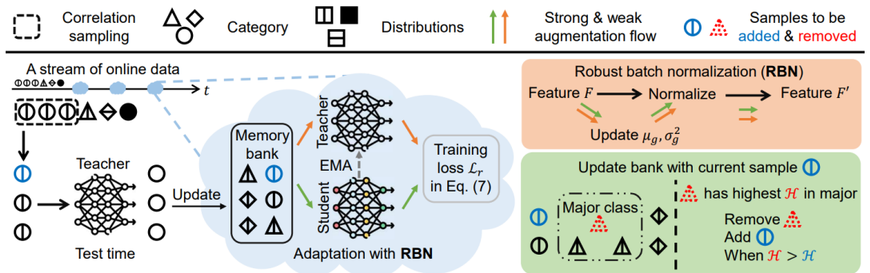
- The assumptions of prior TTA setups, including Fully TTA, Continual TTA and Non-i.i.d. TTA, are often broken in real-world applications.
- In reality, distribution change and data correlation occur simultaneously. For example, in autonomous driving, test samples are highly correlated over time and distribution changes as the surrounding environment alerts.

Robust Test-Time Adaptation (RoTTA).

- Traditional algorithms become more unreliable as the errors accumulate.
- Incorrect estimation in the batch normalization statistics leads to erroneous predictions of test samples, consequently resulting in invalid adaptation.
- The model will easily or quickly overfit to the local distribution caused by the correlative sampling.



Approach



Robust Batch Normalization.

- During adaptation, global statistics (μ_g, σ_g^2) are maintained by EMA :

$$\mu_g = (1-\alpha)\mu_g + \alpha\mu_b, \sigma_g^2 = (1-\alpha)\sigma_g^2 + \alpha\sigma_b^2.$$

- During inference, (μ_g, σ_g^2) are adopted to perform normalization directly.

Category-balanced sampling with timeliness and uncertainty.

- Capacity of $\mathcal{M}$ is equally distributed to each class for category balance.
- Timeliness and uncertainty is considered by:

$$\mathcal{H} = \lambda_t \frac{1}{1+\exp(-\mathcal{A}/\mathcal{N})} + \lambda_u \frac{\mathcal{U}}{\log \mathcal{C}}.$$

Robust training with timeliness.

- Training objective at time t:

$$\min_{\theta_{t+1}^S} \mathcal{L}_r = \frac{1}{\Omega}\sum_{i=1}^{\Omega} \mathcal{L}(x_i^{\mathcal{M}}, \mathcal{A}_i; \theta_t^T, \theta_t^S) = \frac{1}{\Omega}\sum_{i=1}^{\Omega} E(\mathcal{A}_i)\ell(x_i', x_i'')$$

$$E(\mathcal{A}_i) = \frac{\exp(-\mathcal{A}_i/\mathcal{N})}{1+\exp(-\mathcal{A}/\mathcal{N})}, \qquad \ell(x_i', x_i'') = -\frac{1}{\mathcal{C}}\sum_{c=1}^{\mathcal{C}} p_T(c|x_i')\log(p_S(c|x_i''))$$

- Update teacher model by EMA: $\theta_{t+1}^T = (1-\nu)\theta_t^T + \nu\theta_{t+1}^S$.

Experiments

Table 2. Average classification error of the task CIFAR10 → CIFAR10-C while continually adapting to different corruptions at the highest severity 5 with correlatively sampled test stream under the proposed setup PTTA.

Method	motion	snow	fog	shot	defocus	contrast	zoom	brightness	frost	elastic	glass	gaussian	pixelate	jpeg	impulse	Avg.
Source	34.8	25.1	26.0	65.7	46.9	46.7	42.0	9.3	41.3	26.6	54.3	72.3	58.5	30.3	72.9	43.5
BN [53]	73.2	73.4	72.7	77.2	73.7	72.5	72.9	71.0	74.1	77.7	80.0	76.9	75.5	78.3	79.0	75.2
PL [39]	73.9	75.0	75.6	81.0	79.9	80.6	82.0	83.2	85.3	87.3	88.3	87.5	87.5	87.5	88.2	82.9
TENT [70]	74.3	77.4	80.1	86.2	86.7	87.3	87.9	87.4	88.2	89.0	89.2	89.0	88.3	89.7	89.2	86.0
LAME [5]	29.5	**19.0**	20.3	65.3	42.4	43.4	36.8	**5.4**	37.2	**18.6**	51.2	73.2	57.0	**22.6**	71.3	39.5
CoTTA [73]	77.1	80.6	83.1	84.4	83.9	84.2	83.1	82.6	84.4	84.2	84.5	84.6	82.7	83.8	84.9	83.2
NOTE [19]	**18.0**	22.1	20.6	35.6	26.9	**13.6**	26.5	17.3	27.2	37.0	48.3	38.8	42.6	41.9	49.7	31.1
RoTTA	18.1	21.3	**18.8**	**33.6**	**23.6**	16.5	**15.1**	11.2	**21.9**	30.7	**39.6**	**26.8**	**33.7**	27.8	**39.5**	**25.2** (±5.9)

Table 3. Average classification error of the task CIFAR100 → CIFAR100-C while continually adapting to different corruptions at the highest severity 5 with correlatively sampled test stream under the proposed setup PTTA.

Method	motion	snow	fog	shot	defocus	contrast	zoom	brightness	frost	elastic	glass	gaussian	pixelate	jpeg	impulse	Avg.
Source	30.8	39.5	50.3	68.0	29.3	55.1	28.8	29.5	45.8	37.2	54.1	73.0	74.7	41.2	39.4	46.4
BN [53]	48.5	54.0	58.9	56.2	46.4	48.0	47.0	45.4	52.9	53.4	57.1	58.2	51.7	57.1	58.8	52.9
PL [39]	50.6	62.1	73.9	87.8	90.8	96.0	94.8	96.4	97.4	97.2	97.4	97.4	97.3	97.4	97.4	88.9
TENT [70]	53.3	77.6	93.0	96.5	96.7	97.5	97.1	97.5	97.3	97.2	97.1	97.7	97.6	98.0	98.3	92.8
LAME [5]	**22.4**	**30.4**	43.9	66.3	**21.3**	51.7	**20.6**	**21.8**	39.5	**28.0**	48.7	72.8	74.6	**33.1**	**32.3**	40.5
CoTTA [73]	49.2	52.7	56.8	53.0	48.7	51.7	49.4	48.7	52.5	52.2	54.3	54.9	49.6	53.4	56.2	52.2
NOTE [19]	45.7	53.0	58.2	65.6	54.2	52.0	59.8	63.5	74.8	91.8	98.1	98.3	96.8	97.0	98.2	73.8
RoTTA	31.8	36.7	**40.9**	**42.1**	30.0	**33.6**	27.9	25.4	**32.3**	34.0	**38.8**	**38.7**	**31.3**	38.0	42.9	**35.0** (±5.5)

Table 4. Average classification error of DomainNet while continually adapting to different domains with correlatively sampled test stream

Source	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	83.9	65.4	88.6	48.0	59.1	69.0
inf	61.8	N/A	66.9	96.0	50.0	70.6	69.1
pnt	56.5	83.7	N/A	94.2	42.6	63.4	68.1
qdr	89.2	99.0	98.6	N/A	95.0	92.3	94.8
rel	49.4	80.4	51.5	93.4	N/A	63.3	67.6
skt	47.5	88.2	62.9	87.1	51.8	N/A	67.5
Avg.	60.9	87.0	69.1	91.9	57.5	69.7	72.7

BN	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	88.6	70.7	90.5	65.4	67.0	76.5
inf	68.6	N/A	74.2	96.2	69.9	76.8	77.1
pnt	60.8	87.9	N/A	94.3	62.3	68.7	74.8
qdr	80.3	97.7	92.6	N/A	88.7	88.1	89.5
rel	57.9	87.1	63.1	94.3	N/A	70.8	74.6
skt	50.4	87.6	64.6	89.6	63.1	N/A	71.1
Avg.	63.6	89.8	73.0	93.0	69.9	74.3	77.3

PL	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	94.5	98.9	99.5	99.7	99.7	98.5
inf	82.6	N/A	99.2	99.6	99.7	99.3	96.1
pnt	78.6	99.4	N/A	99.7	99.6	99.7	95.4
qdr	81.7	99.5	99.6	N/A	99.7	99.8	96.1
rel	73.5	99.4	99.2	99.6	N/A	99.7	94.3
skt	64.8	99.2	99.4	99.7	99.7	N/A	92.6
Avg.	76.2	98.4	99.3	99.6	99.7	99.6	95.5

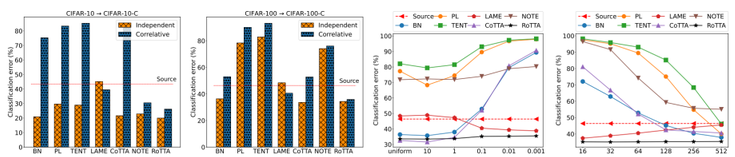
TENT	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	87.5	71.9	94.2	96.2	98.9	89.7
inf	68.6	N/A	75.0	97.3	95.9	98.7	87.1
pnt	61.7	87.1	N/A	96.4	95.3	98.8	87.8
qdr	78.9	97.1	91.6	N/A	89.2	88.7	89.1
rel	57.8	86.4	68.1	96.9	N/A	96.7	81.2
skt	51.9	87.2	69.1	95.3	97.3	N/A	80.1
Avg.	63.8	89.0	75.1	96.0	94.8	96.4	85.8

LAME	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	82.2	64.5	87.7	46.9	58.9	68.0
inf	60.1	N/A	65.7	95.4	48.5	69.4	67.8
pnt	55.8	81.5	N/A	93.3	41.3	62.1	66.8
qdr	88.3	99.1	99.0	N/A	94.9	92.2	94.7
rel	48.0	79.3	50.1	91.6	N/A	60.2	65.8
skt	45.6	87.1	59.5	83.9	49.9	N/A	65.2
Avg.	59.6	85.8	67.8	90.4	56.3	68.6	71.4

COTTA	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	90.6	77.9	89.3	76.3	72.7	81.4
inf	74.5	N/A	82.0	95.7	80.2	81.5	82.8
pnt	66.3	89.8	N/A	93.4	74.0	75.4	79.8
qdr	82.3	98.2	94.6	N/A	92.5	90.1	91.5
rel	64.0	90.3	73.2	93.5	N/A	77.6	79.7
skt	56.1	89.2	71.9	89.2	73.5	N/A	76.0
Avg.	68.6	91.6	79.9	92.2	79.3	79.5	81.9

NOTE	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	89.2	73.0	94.8	98.4	99.4	91.0
inf	75.4	N/A	78.7	98.7	98.1	99.5	90.1
pnt	64.7	89.8	N/A	97.8	98.4	99.2	90.0
qdr	74.7	97.2	92.2	N/A	93.5	99.6	91.4
rel	61.3	89.2	68.9	98.8	N/A	99.2	83.5
skt	55.2	89.7	70.1	96.9	98.3	N/A	82.0
Avg.	66.3	91.0	76.6	97.4	97.3	99.4	88.0

RoTTA	clp	inf	pnt	qdr	rel	skt	Avg.
clp	N/A	85.5	62.0	82.0	49.3	59.8	67.7
inf	61.8	N/A	63.7	91.5	52.5	67.6	67.4
pnt	53.3	84.1	N/A	89.1	47.3	61.4	67.0
qdr	77.5	97.0	89.8	N/A	80.3	82.2	85.3
rel	49.1	82.3	50.3	88.0	N/A	61.1	66.2
skt	42.6	83.7	54.4	80.9	47.5	N/A	61.8
Avg.	56.8	86.5	64.0	86.3	55.4	66.4	**69.2** (±3.2)



(a) CIFAR10-C. (b) CIFAR100-C. (c) δ. (d) Batch size.

Table 5. Classification error of different variants of our RoTTA.

Variant	CIFAR10-C	CIFAR100-C	Avg.
RoTTA w/o RBN	75.4	51.3	63.4
RoTTA w/o CSTU	47.1	46.3	46.7
RoTTA w/o RT	78.2	95.0	81.6
RoTTA	**25.2**	**35.0**	**30.1**

Method	CIFAR10-C	CIFAR100-C	Avg.
Source	43.5	46.4	46.9
BN [53]	75.2	52.9	64.1
PL [39]	75.2	52.9	60.1
TENT [70]	82.3	93.2	87.8
LAME [5]	39.5	40.6	40.1
NOTE [19]	30.5	76.1	53.3
CoTTA [73]	83.1	52.8	67.9
RoTTA	**26.2** (±4.3)	**35.9** (±4.7)	**31.1** (±9.0)

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